NEUROLOGICAL EVALUATION OF CHILDREN WITH LEARNING DISORDERS

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SCHAIN, R. J.: *Neurological evaluation of children with learning disorders.* Neuropädiatrie 1 : 307—317 (1970). Neurological findings in 40 children referred with the principal of complaint of school learning problems are reviewed. Speech disorders and hyperkinetic syndromes were the most common features of the anamnesis. The final diagnosis in these subjects were: Minimal Brain Dysfunction (23), Behavior Disorder of Childhood (11), Mental Retardation (4), Acute Infantile Hemiplegia (1), High Myopia (1). Seizures disorders and borderline cerebral palsy syndromes are the most important neurological conditions to be excluded. It is believed that the distinction between minimal brain dysfunction syndromes and behavior disorders of childhood cannot be reliably based upon soft neurological signs, borderline EEG abnormalities or evidence of perceptual disturbances on psychological testing.

Learning disorders Minimal brain dysfunction Neurological evaluation

Introduction

The referral of children for neurological evaluation with the principal complaint of school learning difficulties is occurring with increasing frequency in pediatric neurology facilities. These children are sent to a neurologist because of a suspicion by parent, teacher, psychologist, or physician that the child's learning difficulties are of neurological origin. An extensive literature on "minimal brain dysfunction" (or similar terms) as a diagnostic concept applicable to many of these children has appeared in recent years (WORK, H. H., and HALDANE, J. E., 1966; PINCUS, J. H., and GLASER, G. J., 1966; SCHAIN, R. J., 1968). This expression might be briefly defined as a label attached to children of normal intelligence with developmental disorders of attentiveness (hyperkinesis), language (speech delays) and perceptual motor functions (learning disorders). It is generally agreed that this expression is descriptive of symptoms with multiple determinants and should not be regarded as a unitary disorder (BIRCH, H. G., 1964).

In developing a neurological approach toward this problem, it must be remembered that minimal brain dys-

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function is a formulation of clinicans. School age children are brought for neurological evaluation because of poor achievement in the classroom. It is the responsibility of the neurologist to construct a differential diagnosis based upon the presenting complaint and to rule out potential neurological disorders. The purpose of this report is to describe clinical findings in 40 children referred to a pediatric neurology facility with the chief complaint of learning problems and to outline a differential diagnosis in these circumstances. These cases consisted of approximately 20% of all referrals during this time period and were exceeded in frequency only by seizure disorders indicating the common occurrence of this referral.

Methods

These 40 children were referred over a two year period because of an impression by a responsible adult that the child's school difficulties might be of neurological origin. The patients ranged in age from 7 to 13 years of age with a mean age of 9.3 years. There were 27 males and 13 females. All of the group originated from middle to high income homes. Two were negro and 38 were white. Histories were obtained and neurological examinations were carried out by the author. EEG studies and skull x-rays were performed on 20 subjects when there was a suggestion of abrupt or progressive behavioral changes, evidence of focal neurological signs or a past history of seizures. Psychometric data was available on 22 of the 40 subjects.

Results

Anamnesis: Pertinent historical information is summarized in Table I.

The high incidence of speech disorders and hyperkinetic behavior is notable in these subjects as in most other studies. Hyperkinesis was judged to have been present if a parent believed restless, impulsive behavior significantly interfered with the child's adjustment at home or at school. It has become recognized that hyperkinetic behavior is not primarily a function of an increase of quantitative activity but rather with a short attention span and motor impulsivity which interferes with normal interaction of the child with his environment (SCHULMAN, J. L., et al., 1965; EISENBERG, L., 1964).

Speech disturbances were usually described as delayed or indistinct speech. These speech disorders are perhaps best regarded as examples of articulatory apraxia as described by

Table I Incidence of Historical Events

Speech Disorders	17
Hyperkinetic syndromes	17
Low birth weight (under 5 lb.–8 oz.)	9
Strabismus	5
Single seizures	3
Major perinatal abnormalities	3
Recurrent headaches	2
Cerebrovascular accidents	1
Meningitis	1

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INGRAM and REID (INGRAM, T. T. S., and REID, J. F., 1956). Many of these children did not manifest evidence of speech or motility disturbances at the time of examination. The parents stated that these symptoms were present at an earlier ago but were no longer troublesome.

Eight of the 9 low birth weight infants were between 4 and 5 1/2 lbs., at the time of birth. The remaining patient with a birth weight of 3 lbs., 14 oz., was the child who received a primary diagnosis of high myopia. Five of the children manifested strabismus requiring ophthalmological attention. In no case was the strabismus felt to be directly responsible for the learning difficulties. This is in accord with prevailing ophthalmological views that the disturbances of conjugate eye movements are not responsible for reading disorders (GOLDBERG, H. K., and DRASH, P. W., 1968).

Examination: Neurologic examination resulted in evidence of mild, previously unrecognized hemiparesis in one child manifest by reduced muscle mass and weakness of dorsiflexion of the foot. This was the residual of an acute infantile hemiplegia (undiagnosed) occurring at 10 months of age. This was the only child in which a clearcut neurological disorder was evident on physical examination.

Ten of the children manifested evidence of speech disturbances at the time of examination. These were characterized by indistinct speech, perseveration and occasional alteration of word order which were not severe enough to significantly interfere with communication and were not regarded as major dysarthric disorders.

Major EEG abnormalities consisting of paroxysmal polyspikes or spikes and wave discharges were present in 4 patients (Figure 1). One of these children was hemiparetic and another was subject to recurrent headaches.

None had clinical evidence of seizures at the time of examination. In addition, a variety of minor dysrhythmic changes (diffuse slowing, posterior slowing, poor organization) were often present but of questionable significance.

Psychometric data were available on 22 of the patients. The WECHSLER Intelligence Scale for Children (WISC) or the Stanford Binet was administered to all of the children tested. In this study, the major use of the psychometric data was to confirm a diagnosis of global mental retardation in four of the group (IQ 42–78). In all other subjects, the fullsacle intelligence quotient ranged from 82 to 123.

Diagnosis: The final diagnoses in these 40 subjects are tabulated in Table II.

It should be noted that only one of these children manifested evidence of an overt cerebral palsy syndrome. One

Table II Final Diagnoses in 40 children with the principal complaint of school learning problems (ages 7–13 years)

Minimal Brain Dysfunction	23
Behavior Disorder of Childhood	11
Mental Retardation	4
Acute Infantile Hemiplegia	1
High Myopia	1

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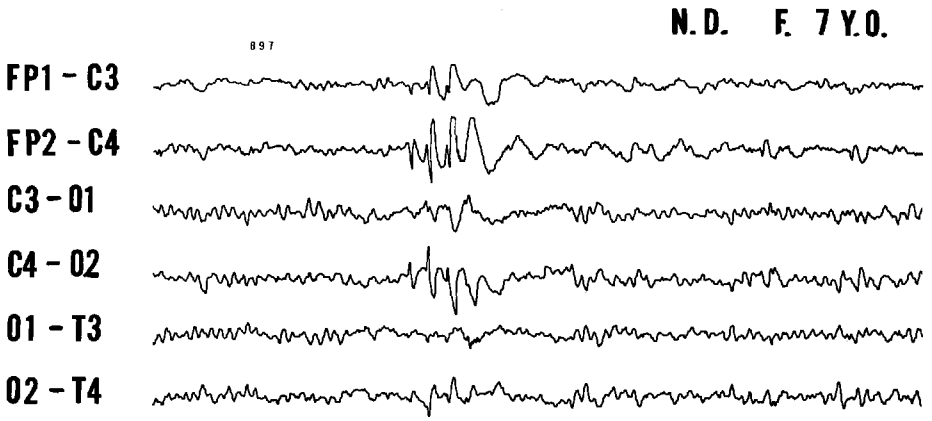


Figure 1 Seven year old girl with the principal complaint of learning difficulties in school. No history of clinical seizures. Focal spike and wave discharges are seen appearing over the right central region

child suffered with a high myopia of 20 diopters that was responsible for her learning problems. Four of the children fell into the category of global mental retardation based upon evidence of significant delays in motor, speech, and social functions and confirmed by psychometric testing.

The remainder of the 34 patients were divided into two diagnostic categories: minimal brain dysfunction or behavior disorders in childhood.

The concept of minimal brain dysfunction has often been criticized because of the poorly defined nature of the syndrome and the lack of reliably demonstrable neurological abnormalities. Nevertheless, this concept has been felt to be essential in the categorization of children with learning disorders (SCHAIN, R. J., 1968). In this study, this diagnosis has been applied to children with developmental disorders in the areas of language, attentiveness, and learning in whom overall intelligence is within normal limits. Delayed establishment of laterality until 5 to 6 years of age was commonly noted by parents. Family histories of similar symptoms of siblings, parents, or near relatives could frequently be elicited. Greater precision in the delineation of this syndrome does not appear to be possible at the present time.

The diagnosis of behavior disorder of childhood was made in those children with major behavioral problems

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Table III Correlation of selected historical and physical findings with major final diagnosis (numbers refer to number of cases)

	Minimal Brain Dysfunction 23 (14 M–9 F)	Behavior Disorder of Childhood 11 (9 M–2 F)	Mental Retardation 4 (3 M–1 F)	Acute Infantile Hemiplegia 1 (1 M)	High Myopia 1 (1 F)
Speech Disorders	8	4	4	1	
Hyperkinetic Syndrome	9	6	2		
Low Birth Weight	7	1			1
Perinatal Disturbances	2		1		
Paroxysmal EEG Abnormalities	1	1	1	1	
Strabismus	1	1	2		1
Recurrent headaches	2				

that appeared to be based upon more than school difficulties. Family disturbances that were obviously affecting the child were often present. The behavior problems were of a serious nature such as threatening a parent with a knife or expressing suicidal impulses. Stealing was a complaint of parents of 5 of the 11 children diagnosed in this manner. It was clear, however, that children categorized as minimal brain dysfunction also had behavior problems although of lesser degree. The issues involved in this distinction are further discussed below.

Correlation of certain historical and physical findings with the final diagnosis is given in Table III.

It is evident that speech disturbances and hyperkinesis are evenly distributed between the groups diagnosed as behavior disorder of childhood and minimal brain dysfunction. Sex ratio is more heavily weighted in favor of males in the behavior disorder group (4.5 : 1) than in the minimal brain dysfunction group (1.6 : 1). The high incidence of low birth weight is notable in the minimal brain dysfunction group.

Discussion

The low incidence of overt neurological abnormalities in children with learning disorders specifically referred for neurological consultation is strikingly evident. This will come as no surprise to those familiar with the characteristics of children with learning problems. Referral to a neurologist is usually motivated more by confusion and anxiety regarding an underachieving child than by concern over a neurological symptom.

It is to be emphasized that this analysis refers to childen referred with the *principal* complaint of school learning problems. It is well known that

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Table IV Neurologic Differential Diagnosis of Learning Disorders

Conditions	Major Findings
Seizure disorders	Prolonged or paroxysmal behavioral disturbances (GOLDENSOHN, E. S., and GOLD, A. P., 1960)
	Bilateral spike-wave discharges or temporal foci (GLASER, G. H., and GOLUB, L. M., 1955)
Cerebral palsy syndromes (borderline) (INGRAM, T. T. S., 1964)	Hemiparesis Ataxia Dyskinesias Diplegia
Hemispheric brain tumors	Focal neurological and EEG abnormalities Abnormal neuroradiological contrast studies
Subacute sclerosing panencephalitis	Myoclonic seizures Rhythmic diffuse wave and spike complexes in EEG (BRAIN, W. R., et al., 1948)
Juvenile cerebromacular degeneration	Loss of visual acuity Speech disturbances Paroxysmal EEG changes (ELLINGSON, R. J., and SCHAIN, R. J., 1969)
HUNTINGTON's Chorea	Positive family history Rigidity and involuntary movements (MARKHAM, C. H., and KNOX, J. W., 1965)

learning difficulties may be associated with a wide variety of central nervous system disorders. The more important of these have been outlined in Table IV. The presence of structural brain disease is usually recognized in most of these disorders and the presenting complaints are rarely limited to learning problems. However, disturbances of mentation and behavior may occasionally be the sole manifestation of major neurological disorders. Any evidence of recent alterations of behavior or of unusual progression of symptoms should be regarded with suspicion. The resemblance of the symptoms of GERSTMANN's syndrome[1] with some of the problems of children with learning disorders (HERMANN, K., and NORRIE, E., 1958) may serve as a reminder that spatial and sequential disorientations interfering with school performance may result from cerebral lesions.

The occurrence of EEGs with major epileptiform abnormalities is not an uncommon finding in children with minimal brain dysfunction (PAINE, R. S., 1962; CAPUTE, A. J., et al., 1968) (5–10%). This was also true in this study as 4 of the cases manifested paroxysmal disturbances in their EEG (Table III). PAINE regarded this finding as a borderline manifestation of a seizure disorder (PAINE, R. S., 1962).

[1] Gerstmann's syndrome — Finger agnosia, right-left disorientation, dysgraphia and dyscalculia associated with a tumor involving the angular gyrus of the parietal lobe of the dominant hemisphere (Gerstmann, J., 1940).

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The detection of this type of EEG pattern indicates the need for a careful analysis of the child's behavior focussing on a search for seizure phenomena such as brief alterations of consciousness, lipsmacking, chewing movements, disorientations or other behavioral automatisms (GOLDENSOHN, E. S., and GOLD, A. P., 1960; GLASER, G. H., and DIXON, M. S., 1956). However, clinical evidence of seizure states are rarely found in these children, even those with paroxysmal EEG disturbances. The significance of EEG findings of this type in children with learning disorders is yet to be determined but may well adversely affect cognitive functions.

Having excluded major neurological syndroms, the pediatric neurologist is faced with the task of formulating a diagnosis and set of recommendations for the bulk of children who do not fit into well-defined syndromes. The recognition of borderline mental retardation, regardless of etiology, is important because of its educational and social implications. Ten percent of the present series fell into the category of mental retardation. Reliance on abbreviated office testing for diagnostic appraisal is often quite deceptive. The developmental history may give a better index as to overall mental functions, but must be supplemented by formal psychometric testing whenever a question of mental retardation arises. The WISC or Stanford Binet are usually adequate for this purpose.

The most difficult aspect of an evaluation consists of distinguishing between minimal brain dysfunction syndromes and behavior disorders of childhood. This corresponds to the categories of primary reading retardation (developmental dyslexia) and secondary reading retardation (exogenous factors such as anxiety or poor schooling) developed by RABINOWITCH and coworkers (RABINOWITCH, R. D., et al., 1954). "Soft" neurological signs such as clumsiness, choreiform movements, dysdiadochokinesis, mixed laterality, synkinesias, motor impersistence and others have been regarded as evidence of subtle disturbances of brain function by many authors (SCHAIN, R. J., 1968). This concept was first advanced by BENDER in her studies of severely disturbed children diagnosed as childhood schizophrenia (BENDER, L., 1947). Borderline EEG abnormalities are often used to reinforce this concept. From the psychometric side, scatter in subtest scores of the WISC or Stanford Binet and evidence of spatial confusions in the BENDER Gestalt or Goodenough Draw A Person test have been interpreted to indicate cerebral damage. However, careful analysis of the use of most of these approaches reveal that they cannot be relied upon to distinguish poor performance due to adverse environmental factors and cerebral dysfunction of endogenous origin (SCHULMAN, J. E., et al., 1965; HERBERT, M., 1964; PAINE, R. S., et al., 1968). Normative data for most of the soft neurological signs is lacking. The standard intelligence tests such as the WISC or Stanford Binet were originally developed for the measure-

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ment of general intellectual functioning. They are still best utilized as diagnostic instruments for this purpose.

Definite evidence of early developmental disturbances might be regarded as more indicative of cerebral dysfunction (or maturational lag) than neurological abnormalities of dubious reliability. In this regard, it is of interest that speech disorders and hyperkinesis appears with approximately equal frequency in both the behavior disorder of childhood and minimal brain dysfunction groups. It must be concluded that the distinction between these two categories in children with learning disorders is largely based upon the presence or absence of serious behavior problems in the affected child.

All studies of children with behavior disorders indicate a male preponderance of 3 to 4 : 1 or higher (BAHN, A. et al., 1961; BENTZEN, F., 1963). In this study, in which an effort was made to separate psychogenic behavior problems from minimal brain dysfunction, the incidence was only 1.6 : 1 in favor of males in the latter group as compared to 4.5 :1 in the presumed psychogenic group. HALLGREN's (HALLGREN, B., 1950) monumental study in this area reported investigations on 112 cases of "specific dyslexia" and 160 secondary cases (sibs and parents of probands). This is virtually the only report in which test procedures were administered to entire families. The incidence of dyslexia in males, as usual, was higher than in females in primary cases but was almost equally distributed among the sexes in the secondary cases. HALLGREN concluded that factors extraneous to genetically determined learning difficulties may explain the male preponderance. It may be that behavioral reactions to the frustrations produced by learning difficulties are the sex-linked aspect of the problem.

The terminology of minimal brain dysfunction is a continuing problem. Table V summarizes the more historically important terms that may be regarded as equivalent. The evidence that dyslexia and its synonyms as commonly used refer to the same population as those labelled by others as minimal brain dysfunction syndromes has been previously reviewed (SCHAIN, R. J., 1968). Most recently, MACKEITH (MACKEITH, R., 1968) has proposed the expression Neurodevelopmental Disorders as a generic term to characterize these disorders. This is an espe-

Table V Some Equivalents of the Minimal Brain Dysfunction Syndrome

Minimal Cerebral Dysfunction	Developmental Gerstmann's syndrome
Minimal Chronic Brain Syndromes	Developmental aphasia
Minimal Brain Damage	Specific Reading Disability
Clumsy Child Syndrome	Congenital Word Blindness
Visual Motor Disability	Specific Language Disability
Hyperkinetic Behavior Syndrome	Strephosymbolia
Neurodevelopmental Disorders	Developmental Dyslexia

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cially appropriate term, which, to paraphrase an expression of HALLGREN quoted by SILVER and HAGIN (1960) in another context, is "suitable obscure, sufficiently descriptive and devoid of theoretical implications".

Socioeconomic factors have not been included in this discussion because of the particular origins of the study group. Children from depressed urban or rural areas present a major problem in consideration of the epidemiology of learning problems (EISENBERG, L., 1966). Any comprehensive approach to children with learning disorders must include cultural and educational factors in addition to the issues discussed above.

The management of children with learning disorders is a function of the final diagnostic formulation. This requires co-operation with a variety of other professionals, especially in the area of special education and family counselling. Neurological aspects of therapy will primarily be concerned with the use of drugs in the control of hyperkinesis and improvement of attention span. Barbiturates should not be used as paradoxical reactions in the form of excessive irritability and increased activity frequently occur. Dextroemphetamine (5–30 mgm./day) and Methylphenidate (10–80 mgm./day) have been extensively used to control hyperkinesis (EISENBERG, L., 1966; WERRY, J. S., 1968). These drugs should be administered in the same manner as anticonvulsants by gradually increasing dosage until an adequate response or an adverse reaction occurs.

Most of the children with the hyperkinetic syndrome manifested improvement in this aspect of their behavior at the time of examination. The tendency of hyperkinesis to lessen with age is well known although this is not necessarily true of social adjustment (WERRY, J. S., 1968). One might speculate that distractibility might have interfered with the establishment of learning and social patterns at a critical stage of development during the preschool years. The maximum value of drug treatment of the hyperkinetic syndrome may lie in its use prior to the time when a learning disorder has become evident.

The question of the use of anticonvulsant medications in children with epileptiform EEG abnormalities will frequently arise. The literature is not encouraging as to the usefulness of treatment of the EEG in this manner (PINCUS, J. H., and GLASER, G. J., 1966). However, when definite epileptiform features in the form of polyspikes or spike and wave discharges are present, a trial of primadone or diphenylhydantoin may be warranted.

Finally, the pediatric neurologist will recognize that certain parents have a persistent belief that there should be a medical solution to their child's learning problems. This may be reinforced by a wish to avoid recognition of factors in the home adversely affecting the child. An important purpose of the neurological evaluation is to clearly rule out known neurological disorders in order to provide a basis for subsequent counselling. When ne-

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cessary, the parents should be encouraged to obtain help for the resolution of family difficulties that appear to be a significant factor in their child's learning disorder.

Zusammenfassung

Es wird über die Ergebnisse einer neurologischen Untersuchung bei 40 Kindern berichtet, die primär wegen Lernschwierigkeiten überwiesen worden waren. Sprachstörungen und Hyperaktivität waren in der Anamnese am häufigsten zu finden. Bei einem Kind lag außerdem eine Hemiparese vor. Bei 4 der untersuchten Kinder konnten deutlich EEG-Veränderungen gefunden werden, die sich als paroxysmell auftretende Spike-Entladungen manifestierten. Die 40 Kinder konnten der neurologischen Diagnose nach in 5 Gruppen unterteilt werden:

Diskrete Hirnfunktionsstörungen (minimal brain dysfunction syndrome) : 23
Verhaltensstörungen : 11
Retardierte geistige Entwicklung : 4
Akute frühkindliche Hemiplegie : 1
Schwere Myopie : 1

Die Knaben überwogen in der Gruppe der Verhaltensstörungen (4,5 : 1), nicht aber in der Gruppe mit diskreten Störungen der Hirnfunktion (1,6 : 1). In der Tabelle IV wurden die aus neurologischer Sicht wichtigsten differentialdiagnostischen Gesichtspunkte zusammengefaßt, die bei Kindern mit Lernschwierigkeiten mit in Erwägung zu ziehen sind. Krampfleiden und Schwachform eines zerebralparetischen Syndroms sind die wichtigsten neurologischen Krankheiten die bei Kindern mit Lernschwierigkeiten auszuschließen sind. Raumfordernde intrakraniale Prozesse und degenerative Erkrankungen des Zentralnervensystems sollten ebenfalls mit in differentialdiagnostische Überlegungen eingeschlossen werden, obwohl Lernschwierigkeiten eher selten als ein Hauptsymptom solcher Grundkrankheiten beobachtet werden.

Eine zuverlässige Trennung zwischen einer diskreten Hirnfunktionsstörung (minimal brain dysfunction syndrome) und einer Verhaltensstörung ist mit Hilfe „weicher" neurologischer Symptome, mit Grenzbefunden im EEG und mit den Befunden einer Perzeptionsstörung nicht möglich.

Die offensichtliche Knabenwendigkeit der Verhaltensstörungen bei Lernschwierigkeiten könnte durch die besondere Verhaltensreaktion der Knaben begründet sein, mit der diese auf die Lernschwierigkeiten und deren Folgen reagieren. Der Behandlung einer diskreten Hirnfunktionsstörung liegt die pharmakologische Beeinflussung des Hyperaktivitäts-Syndroms zugrunde. Außerdem sollten Kinder mit epileptiformen EEG-Veränderungen antikonvulsiv behandelt werden. Kinder, bei denen eine Störung durch die Situation innerhalb der Familie wahrscheinlich gemacht werden kann, sollten auch von psychologischer Seite Hilfe erfahren.

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